

Kolmogorov's Axioms of Probability

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Introduction

Modern probability theory rests on three axioms proposed by Andrey Kolmogorov in 1933. They turn probability from a collection of intuitions into a rigorous mathematical theory consistent with measure theory. Every theorem about probability – the law of large numbers, the central limit theorem, Bayes' rule – is ultimately a consequence of these three rules and the definitions that accompany them.

Prerequisites

The reader should know what a set is and should be comfortable with the basic set operations (union, intersection, complement).

Theory

A **probability space** is the triple $(\Omega, \mathcal{F}, P)$, where:

- Ω is the **sample space**, the set of all possible outcomes.
- $\mathcal{F}$ is a **sigma-algebra** of subsets of Ω – the collection of events to which we assign probabilities. It is closed under complements and countable unions.
- P is the **probability measure**, a function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying Kolmogorov's three axioms:
 1. **Non-negativity**: $P(A) \geq 0$ for every $A \in \mathcal{F}$.
 2. **Normalisation**: $P(\Omega) = 1$.
 3. **Countable additivity**: for any sequence of pairwise disjoint events $A_1, A_2, \dots$ in $\mathcal{F}$, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

From these axioms every familiar property follows:

- $P(\emptyset) = 0$.
- $P(A^c) = 1 - P(A)$ (complement rule).
- If $A \subseteq B$ then $P(A) \leq P(B)$ (monotonicity).
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (inclusion-exclusion).

The axioms are the minimal, sufficient rules. Violating any of them produces a theory that is either inconsistent or fails to capture behaviour real data exhibit.

Assumptions

The axioms themselves are assumptions; they cannot be derived from more primitive principles. Their justification is empirical: they match the long-run frequencies observed in real random experiments.

R Implementation

We can verify the axioms empirically by simulation for a simple probability space.

```
set.seed(2026)

# Fair six-sided die: Omega = {1, ..., 6}
rolls <- sample(1:6, size = 1e5, replace = TRUE)
```

```

# Axiom 1: every event probability in [0, 1]
empirical_p <- prop.table(table(rolls))
all(empirical_p >= 0 & empirical_p <= 1)

# Axiom 2: probability of the whole sample space is 1
mean(rolls %in% 1:6)

# Axiom 3: additivity for disjoint events
A <- rolls %in% c(1, 2) # "low" outcomes
B <- rolls %in% c(3, 4) # "mid" outcomes
# A and B are disjoint; P(A or B) should equal P(A) + P(B)
c(p_union = mean(A | B),
  p_A_plus_p_B = mean(A) + mean(B))

# Derived property: complement rule
c(p_not_A = mean(!A),
  one_minus_p_A = 1 - mean(A))

# Inclusion-exclusion for non-disjoint events
C <- rolls %in% c(2, 3, 4) # overlaps with A
c(p_union = mean(A | C),
  formula = mean(A) + mean(C) - mean(A & C))

```

Output & Results

```

[1] TRUE          # all empirical probabilities in [0, 1]

[1] 1             # P(Omega) = 1

p_union p_A_plus_p_B
0.6661   0.6661

p_not_A one_minus_p_A
0.6669   0.6669

p_union formula
0.6669   0.6669

```

Empirical frequencies converge to axiomatic probabilities as n grows; the axioms are satisfied by the simulated frequencies.

Interpretation

The axioms are not something one “reports” in a paper; they are the framework within which every probabilistic claim is made. When a manuscript states $P(\text{event}) = 0.23$, it assumes a probability space has been defined implicitly and that P is a valid measure on it.

Practical Tips

- In applied work the sample space is usually obvious (outcomes of a trial, categories of a classification), and the sigma-algebra is the full power set. Measure-theoretic niceties matter only in continuous or infinite settings.
- Countable additivity is strictly stronger than finite additivity; the two agree for finite sample spaces.

- Empirically estimated probabilities (sample proportions) satisfy the axioms by construction, but converge to the true probability only in the limit (law of large numbers).
- When combining events, always check disjointness before adding probabilities; use inclusion-exclusion when events overlap.
- The axioms say nothing about *interpretation* of probability (frequentist, Bayesian, logical); they are compatible with all three.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots